COURSE DESCRIPTION			
Program	**Computer Application & Business Management**		
Course Title	**Business Mathematics**		
Course Code	**2001E**	**Semester**	**II**
Type of Course	**Theory**	**Contact Hours**	**60**
Teaching Scheme (L: T:P:R)	**3:1:0:0**	**Credits**	**4**
CIE Marks	**40**	**ESE Marks**	**60**

Introduction

Business Mathematics is a foundational course designed to equip diploma students with essential mathematical tools required for effective business decision-making. The course integrates mathematical concepts with real-world business applications such as financial analysis, risk assessment, data organization, and economic decision-making. Emphasis is placed on problem-solving, logical reasoning, and interpretation of results in managerial and commercial contexts, ensuring relevance to modern business environments.

Course Objectives

1	To develop numerical and analytical skills required for business decision-making
2	To apply mathematical concepts to real-life business and economic situations
3	To build financial literacy relevant to banking, trade, and industry
4	To enable students to interpret data and solve business problems logically

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic Mathematics			Secondary Education

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Explain and apply matrix operations and determinant concepts to organize and solve structured business data problems	Apply
CO2	Explain and apply basic probability concepts to analyze business situations involving risk and uncertainty	Apply
CO3	Compute and apply financial mathematics concepts related to interest, profit, loss, time value of money, and basic investment decisions	Apply
CO4	Apply basic differentiation and integration techniques to solve simple business and economic problems such as marginal analysis and optimization	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	2	-	-	-	-	-	2	-
CO2	3	3	2	2	-	2	-	-	-	2	-
CO3	3	3	3	2	-	2	-	-	-	3	-
CO4	3	3	2	2	-	2	-	-	-	2	-
Total	12	12	9	8	-	6	-	-	-	9	-
Correlation Level	3	3	2.25	2	-	2	-	-	-	2.25	-
Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"											

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	

3	1	0	0	5.5	60	4	40	60	100	0	0	0	100
L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination													

Syllabus – Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Matrices	Types of matrices - Determinant of a square matrix and properties - Matrix operations: addition, subtraction, scalar multiplication - Matrix multiplication and conformability - Transpose of matrices and related theorems - Applications of matrices in business (cost-output, data organization)	12	4
Module II	Probability	Random experiment, sample space, events - Types of events - Mutually exclusive, equally likely, exhaustive events - Independent and dependent events - Business applications of probability (risk, forecasting, quality control)	10	4
Module III	Financial Mathematics	Simple Interest - Compound Interest - Time Value of Money - Percentage calculations - Profit and Loss - EMI, investment comparison, inflation (introductory)	12	4
Module IV	Calculus & Business Applications	Differentiation - Standard derivatives and rules - Integration of standard forms - Applications in economics and business (marginal analysis, optimization)	11	2

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Introduction to matrices & business relevance	Explain the role of matrices in business	L	CO1	Understand	1
Types of matrices	Identify different types of matrices	L	CO1	Understand	1
Determinant of a square matrix	Compute determinants of matrices	L	CO1	Apply	1
Properties of determinants	Apply determinant properties	L	CO1	Apply	1
Problems on determinant properties	Solve determinant-based problems	T	CO1	Apply	2
Matrix addition	Apply matrix addition	L	CO1	Apply	1
Matrix subtraction & scalar multiplication	Perform scalar operations, Multiply matrices	L	CO1	Apply	2
Conformability for multiplication	Determine conformability	L	CO1	Understand	1
Transpose of a matrix	Find transpose of matrices	L	CO1	Apply	1
Properties of transpose	Apply transpose properties	T	CO1	Apply	2
Numerical problems on transpose	Solve transpose problems	L	CO1	Apply	2
Business applications of matrices	Apply matrices to business data (Case based practice)	L	CO1	Apply	1
Module II					
Random experiment & sample space	Explain basic probability terms	L	CO2	Understand	1
Events and types of events	Classify events	L	CO2	Understand	1
Mutually exclusive events	Apply probability rules by solving problems	L	CO2	Apply	2
Equally likely events	Solve probability problems	T	CO2	Apply	1
Exhaustive events	Identify exhaustive events	L	CO2	Understand	1
Independent events	Explain independent events	L	CO2	Understand	1
Dependent events	Apply probability to dependent events by solving problems	L	CO2	Apply	2
Mixed numerical problems	Solve probability numericals	T	CO2	Apply	2
Probability in quality control	Apply probability in QC cases	L	CO2	Apply	1
Risk and uncertainty in business	Explain business risk	L	CO2	Understand	1

Probability in business decisions	Apply probability to decisions	T	CO2	Apply	1
Module III					
Introduction to financial mathematics	Explain importance of financial math	L	CO3	Understand	1
Simple Interest	Explain Simple Interest components & compute simple interest	L	CO3	Apply	1
Compound Interest	Explain compound interest and compute it	L	CO3	Apply	1
Compound Interest	Apply CI formula to solve problems	T	CO3	Apply	2
Time Value of Money	Explain Time Value of Money and apply calculations	L	CO3	Apply	2
Percentage	Explain percentage concepts & apply percentage calculations	L	CO3	Apply	1
Profit and loss	Explain profit and loss, compute them	L	CO3	Apply	2
Cost price & selling price	Apply cost price – selling price relations	L	CO3	Apply	2
EMI	Explain EMI structure & compute EMI values	L	CO3	Apply	2
Investment comparison	Apply financial math to investments (Case-based learning)	T	CO3	Apply	2
Module IV					
Introduction to calculus in business	Explain the relevance of calculus	L	CO4	Understand	1
Differentiation from first principle	Explain differentiation	L	CO4	Understand	1
Derivatives of standard functions	Find derivatives	L	CO4	Apply	1
Rules of differentiation	Explain differentiation rules	L	CO4	Understand	1
Product rule	Apply product rule	L	CO4	Apply	1
Quotient rule	Apply quotient rule	L	CO4	Apply	1
Mixed differentiation problems	Solve differentiation problems	T	CO4	Apply	2
Introduction to integration	Explain integration concept	L	CO4	Understand	1
Integrals of standard forms	Evaluate integrals	L	CO4	Apply	1
Integration of products & quotients	Apply integration rules	L	CO4	Apply	1
Business meaning of derivatives	Explain marginal concepts (Case-based learning)	L	CO4	Understand	1
Profit maximization	Apply calculus concepts to marginal cost, revenue, and profit	T	CO4	Apply	1
Comprehensive application problems	Solve business calculus problems	L	CO4	Apply	1

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	• Study the topics taught in class • Prepare a concept map on types of matrices with business examples • Or any other activity	5.5
2	• Study the topics taught in class • Solve worksheet on determinant properties using spreadsheet tools • Or any other activity	5.5
3	• Study the topics taught in class • Case-based problems on matrix applications in cost-output analysis. • Or any other activity	5.5
4	• Study the topics taught in class • Apply matrices to simple real-life business problems • Or any other activity	5.5
Module II		

5	• Study the topics taught in class • Prepare glossary of probability terms with real-life business examples • Or any other activity	5.5
6	• Study the topics taught in class • Solve probability problems related to quality control scenarios. • Or any other activity	5.5
7	• Study the topics taught in class • Mini assignment on risk and uncertainty in business decisions • Or any other activity	5.5
8	• Study the topics taught in class • Analyze business risk situations using probability • Or any other activity	5.5
Module III		
9	• Study the topics taught in class • Calculate simple interest for different banking products. • Or any other activity	5.5
10	• Study the topics taught in class • Comparison study of compound interest across banks. • Or any other activity	5.5
11	• Study the topics taught in class • Time value of money problems using EMI calculators • Or any other activity	5.5
12	• Study the topics taught in class • Percentage and profit–loss calculations from retail case studies • Or any other activity	5.5
Module IV		
13	• Study the topics taught in class • Practice problems on differentiation using standard functions • Or any other activity	5.5
14	• Study the topics taught in class • Business case problems on marginal cost and revenue • Or any other activity	5.5
15	• Study the topics taught in class • Optimization problems for profit maximization • Reflective report on use of mathematics in business decision-making • Or any other activity	5.5
Total Self-Learning hours		82.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Business Mathematics	Dorai Rajan & Sundaresan	S. Chand
2	Business Mathematics and Statistics	P.R. Vittal	Margham Publications

REFERENCE			
Sl. No.	Title	Author	Publication
1	Business Mathematics	Sancheti & Kapoor	S. Chand
2	Mathematics for Management	Raghavachari	Tata McGraw-Hill
3	Business Mathematics	V.K. Kapoor	S. Chand

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://nptel.ac.in/courses/110/105/110105095/	I
2	https://www.khanacademy.org/math	II
3	https://openstax.org/subjects/math	III

4	https://ocw.mit.edu	IV

Assessment Methodology – Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60
Converted to						20	20	
Marks	5	15				20		60
Total	40							
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		Total Marks
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60